

Aysan Ghayouri

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SUMMARY

AI Engineer with 3 years of experience in Machine Learning, Deep Learning, and Multimodal AI, with experience building Agentic AI and Generative AI systems, including LLM-powered agents, RAG applications, and Vision-Language Model solutions.

EDUCATION

Iran University of Science and Technology

Tehran, Iran

Master of Science in Artificial Intelligence

University of Tehran

Tehran, Iran

Bachelor of Science in Physics

PROJECTS

Agentic Travel Platform

🌐 GitHub

- Developed a multi-agent travel planning platform using LangGraph and FastMCP, integrating real-time flight and hotel search through **Snapptrip**.
- Designed a multi-agent workflow for a travel assistant, featuring a supervisor for task routing, specialized flight and hotel agents, parallel search execution, and a planner for personalized travel recommendations.
- Built a FastAPI backend with JWT authentication, persistent conversations, and streaming responses using Server-Sent Events (SSE).
- Implemented long-term agent memory using LLM-based preference extraction, PostgreSQL, and pgvector for semantic retrieval, with LangGraph checkpointing for conversation state persistence.
- Containerized the application with Docker Compose and integrated Langfuse for agent observability and tracing.

Shopping Agent

🌐 GitHub

- Built an AI-powered shopping assistant for **Digikala** using the OpenAI Agents SDK and FastMCP to enable tool-based product search, retrieval, and recommendation.
- Implemented conversational memory with Mem0 to support personalized recommendations across sessions.
- Containerized the application with Docker and deployed it on AWS ECS.
- Developed a CI/CD pipeline with GitHub Actions for automated security scanning with Trivy, Docker image builds, and deployment workflows.
- Integrated OpenAI Tracing for agent observability and debugging, and used MCP Inspector for tool inspection and testing.

RAG Chatbot for Document Question Answering

🌐 GitHub

- Developed an end-to-end Retrieval-Augmented Generation (RAG) chatbot for document question answering, including PDF ingestion, chunking, retrieval, reranking, and LLM-based response generation.
- Implemented semantic retrieval using FAISS and transformer embeddings with Cross-Encoder reranking to improve context relevance and answer quality.
- Designed a modular RAG pipeline supporting OpenAI and DeepSeek LLMs with configurable generation backends, and containerized the application using Docker.

Vision-Language Model Evaluation Framework

🌐 GitHub

- Built a modular and reproducible evaluation framework for Vision-Language Models including Qwen2.5-VL and DeepSeek-VL2 across VQA benchmarks.
- Implemented configuration-driven evaluation pipelines for MME, POPE, ScienceQA, and TextVQA to analyze model performance in reasoning, OCR, and hallucination scenarios.
- Developed a model-agnostic inference framework enabling standardized benchmarking and extension to new vision-language models.

EXPERIENCE

AI Engineer

1 year

Part AI Research Center

- Evaluated and benchmarked Vision-Language Models (Qwen-VL, LLaVA, DeepSeek-VL, MedGemma) on multimodal reasoning and visual question answering tasks.
- Adapted foundation models using parameter-efficient methods including LoRA, QLoRA, and prompt learning techniques.
- Developed pipelines for multimodal dataset curation, preprocessing, experimentation, and performance analysis.

PUBLICATIONS

Token-Based Dual-View Fusion and Adaptation of Large Vision Models for Breast Cancer Classification,
arXiv preprint, 2026 · DOI: 10.48550/arXiv.2607.06309 · Code: GitHub

TECHNICAL SKILLS

Programming: Python, SQL

Tools: Git, GitHub, Linux

Machine Learning: PyTorch, TensorFlow, Hugging Face Transformers, NumPy, Scikit-learn

LLM & Agent Frameworks: OpenAI Agents SDK, LangChain, LangGraph, FastMCP, CrewAI, AutoGen

RAG, Memory & Vector Databases: Mem0, Qdrant, FAISS, ChromaDB, pgvector

Backend & Databases: FastAPI, PostgreSQL, SQLAlchemy, Alembic

Observability: Langfuse, OpenAI Tracing, MCP Inspector

Deployment & CI/CD: Docker, Docker Compose, AWS ECS, GitHub Actions

AI Techniques: Prompt & Context Engineering, Tool/Function Calling, Agent Memory, Multi-Agent Systems, LLM Evaluation